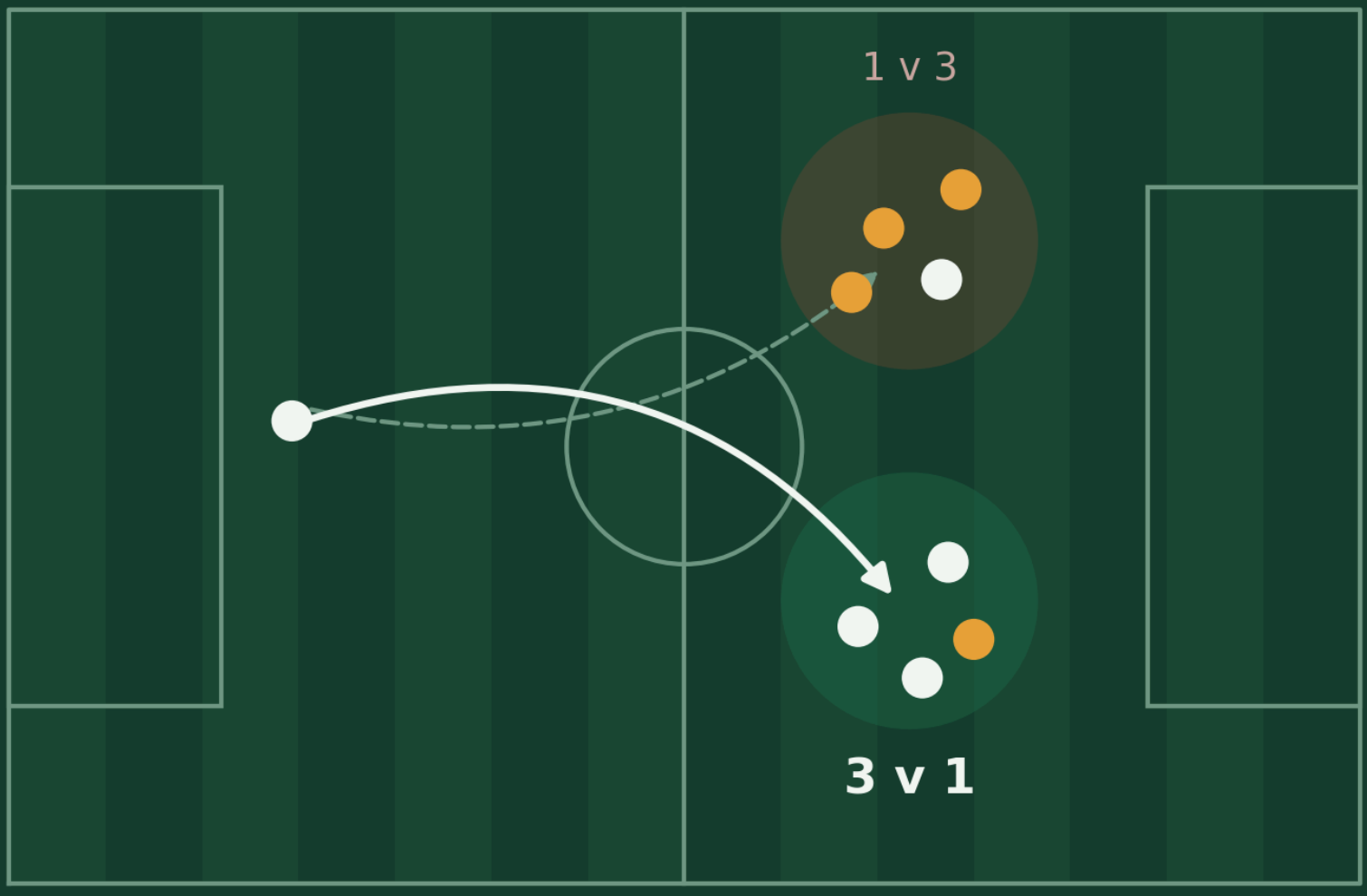
WHITE PAPER · PROJECT UNDERDOG · OCTOBER 2026

FOOTBALL BELIEFS, TESTED.

What actually gives a team an edge? We put 20 coaching beliefs and about 9,400 candidate rules to the test on 3.6 million real decisions and 30 tracked matches, with each test written down before we ran it. Two edges held up fully, three more held up in part, and most of the beliefs failed.



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Independent football analyst

github.com/ranausmanai/count-and-strike

EXECUTIVE SUMMARY

THE SHORT VERSION

Coaches make thousands of in-game decisions on the strength of beliefs: press on the back pass, shoot more, attack their weak side. We tested which of these beliefs give a measurable edge, and which only look right.

3.6M
on-ball decisions analysed

~9,400
ideas and candidate rules tested

2 + 3
edges confirmed, plus partly confirmed

16
popular beliefs with no measurable edge

FINDING	STATUS	EVIDENCE IN ONE LINE
One more where a long ball lands	CONFIRMED	Ball kept 84–89% of the time with +1 locally, vs 63–72% when level. Holds in two independent kinds of data and survives every stress test.
Set pieces as the underdog's equaliser	CONFIRMED	Team quality drives set-piece chances ⅓–⅔ as much as open-play chances, in three datasets.
Bait one side to open the other	PARTLY	The block really slides and the far side opens (all populations). The switch itself shows no extra chances.
Pass into the lane where you outnumber them	PARTLY	Completed 2–4 points more often. No extra chances once the pass is genuinely available.
Count the defenders when you win the ball	PARTLY	Counters are worth 2× more with ≤3 behind the ball. Whether "go fast" beats keeping it is not proven.
"Shoot more"	ILLUSION	The apparent gain was present before the shot. The extra shots a coach would add are worth ≈ 0.
16 common beliefs	NO EDGE	Including "most vulnerable after scoring", pressing triggers, quick restarts, the opponent's "weak side" and early subs.

WHAT THIS MEANS FOR A CLUB

The cheapest edges are in the moments teams treat as low-value: long balls and set pieces. They are also where weaker teams currently lose most. Much time spent on pressing triggers and match-up planning goes to things that show no measurable return.

HOW TO READ THIS PAPER

All results come from observational match data, not controlled trials. "Confirmed" means a rule was found in one population, repeated in two others, and survived stricter checks designed to catch false patterns. It does not guarantee the same effect for every team.

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KEY TERMS

Chance (xG)	Expected goals: the probability that a shot becomes a goal. A chance worth 0.10 xG is scored about 1 time in 10. "Net xG" is our chances minus the opponent's over a time window (usually 20 s).
Population	A group of matches analysed separately: men's international tournaments, men's club football, and women's football.
Confirmed	Found in men's internationals, then repeated in *both* men's club and women's football with the same direction and statistical significance.
Like-for-like	A decision is compared only with alternatives taken in the same situation: same zone, pressure, defenders ahead and teammates ahead.
Stress test	A re-estimate that adjusts for up to 19 details of the moment ("doubly robust" estimation), plus a placebo check (below).
Placebo check	Was the 20 s *before* the decision already different? If so, the "effect" describes a good moment, not a good choice.
Local balance (+1)	Our players minus theirs within 10 m of where a long ball lands.

1 WHY BELIEFS NEED TESTING

Football runs on received wisdom. Much of it is sound. Some of it is a story that fits a few memorable moments. The problem is telling them apart.

RESULTS ARE TOO NOISY TO JUDGE TACTICS

A tactic worth four points a season would take a single club about **57 seasons** of results to prove statistically. Match outcomes simply contain too much luck. That is why tactical beliefs persist whether or not they work.

DECISIONS ARE NOT

A single season contains hundreds of thousands of on-ball decisions. If a choice really helps, it should show up across thousands of comparable moments, in different leagues, in men's and women's football alike.

THE TRAP: GOOD MOMENTS VS GOOD CHOICES

Players are not random. They go forward when there is space, and shoot when the shot is on. So "going forward works" in raw data mostly describes *space*, not a decision. Most published football statistics never separate the two.

THE QUESTION WE STARTED WITH

Project Underdog began with one question: **is there something a weaker team can do, in the moment, that gives it a real edge over a stronger one?** We tested every idea we could find or generate against that standard, and published every failure.

WHAT IS DIFFERENT ABOUT THIS STUDY

- **Pre-registered:** 30+ research rounds and 13 parallel studies, each written to a public git log before its analysis ran.
- **Replicated:** a rule had to hold in three separate populations, not one.
- **Stress-tested:** doubly robust re-estimation and a placebo on the moment before each decision.
- **Self-correcting:** when stricter tests arrived, we killed our own earlier headline findings, and say so here.

2 DATA AND METHOD

SOURCE	COVERAGE	USED FOR
StatsBomb 360 (open data)	418 matches; every visible player's position at each event. Men's international (World Cup 2022, Euro 2020/2024, AFCON 2023); men's club (La Liga, Ligue 1, Bundesliga, MLS); women's (World Cup 2023, Euro 2022/2025).	All situation-based tests
StatsBomb events	1,551 big-five league matches (2015/16), 577 women's and ISL matches	Set pieces, game state, restarts
Wyscout (Pappalardo et al. 2019)	About 1,900 matches, 2017/18	Replication
Tracking	30 matches at 10–25 Hz: DFL/IDSSE Bundesliga (7), SkillCorner A-League (20), Metrica sample (3)	Movement checks, realism targets
Betting odds (football-data.co.uk)	45,000 matches	Team strength, season simulation

EVERY IDEA PASSED THROUGH THE SAME FIVE GATES

1 · REGISTER
The hypothesis, direction and test are written to git before any outcome is seen.

2 · DISCOVER
Tested in men's international football. Many rules at once, so false discoveries are controlled.

3 · REPLICATE
The same direction is required in men's club *and* women's football.

4 · LIKE-FOR-LIKE
Compared only within matched situations, then re-estimated with 7–19 details of the moment.

5 · PLACEBO
The 20 s before the decision must not already differ.

OUTCOME

Usually **net xG over the next 20 seconds** (our chances minus theirs), checked at 45–60 seconds for patience effects. For long balls, the outcome is whether we still have the ball 5 seconds later.

SITUATIONS

Each decision is placed in a cell: pitch zone (6 depths × 3 channels), opponents goal-side, distance to the nearest opponent, and teammates ahead. Choices are compared only within the same cell.

STATISTICS

Benjamini–Hochberg false-discovery control in discovery, Holm correction in confirmation, cross-fitted gradient-boosting nuisance models for doubly robust estimates, and match-clustered standard errors.

FUNNEL

About 9,400 ideas → a few dozen passed discovery → 13 replicated in round 13 → 5 survived the round-22 stress tests in full or in part.

CONFIRMED

3 ONE MORE WHERE IT LANDS

A long ball is rarely won by the player it's aimed at. It's won by whichever side has numbers around the landing spot. Aim at a *place* where you have one more player than them, not at a player.

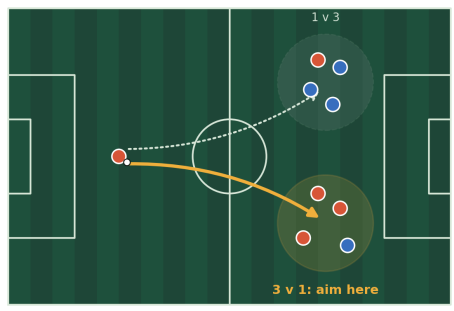


Figure 1. The same kick, two targets: 1 v 3 (top) and 3 v 1 (bottom).

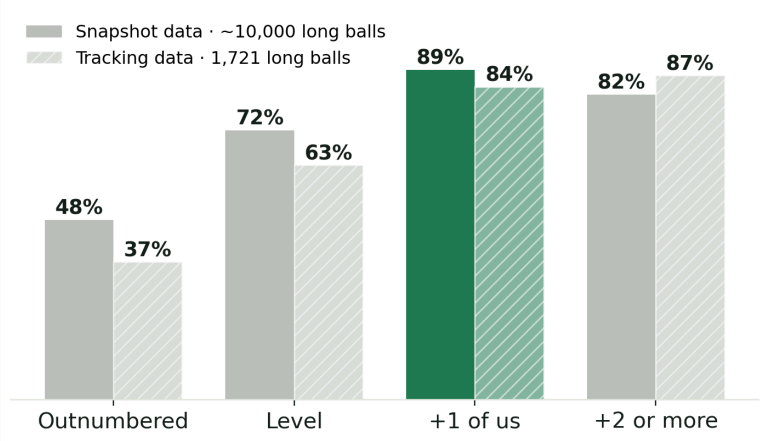


Figure 2. Long balls still in our possession 5 s later, by local balance. Snapshot data (StatsBomb 360, ~10,000 balls) and tracking data (1,721 balls, balance measured on arrival).

TEST	MEN'S INTL	MEN'S CLUB	WOMEN'S	VERDICT
Retention, superiority vs outnumbered (stress-tested, per-pop estimate ± SE)	+25.7 ± 5.3 pts	+20.2 ± 5.7	+18.7 ± 10.2	PASS
Placebo: net xG in the 20 s before the kick	+0.0017 pooled (no difference)			CLEAN
Tracking data replication (Bundesliga, A-League, sample)	37% outnumbered · 63% level · 84% at +1 · 87% at +2			REPLICATES
Net xG over 20 s, superiority vs not	−0.001	−0.017	+0.003	NO EFFECT

IN PRACTICE

On goal kicks and long balls the target is never alone: one named runner arrives beside him so it's 2 v 1. A second extra player added nothing, and committing players did not raise chances conceded within 45 s.

WHERE UNDERDOGS LOSE IT

Underdogs' long balls land with a numbers advantage less often (27–36% vs 34–44% for favourites) and are kept less often (66–75% vs 74–82%).

CAVEAT

This is a possession rule. Keeping the ball did not by itself produce more chances in the following 20–45 s. Its value is in not giving the ball away cheaply.

CONFIRMED

4 SET PIECES: THE EQUALISER

For every source of chances, we measured how strongly it rises with team strength (its "elasticity"). Set pieces are where a better squad's advantage shrinks most.

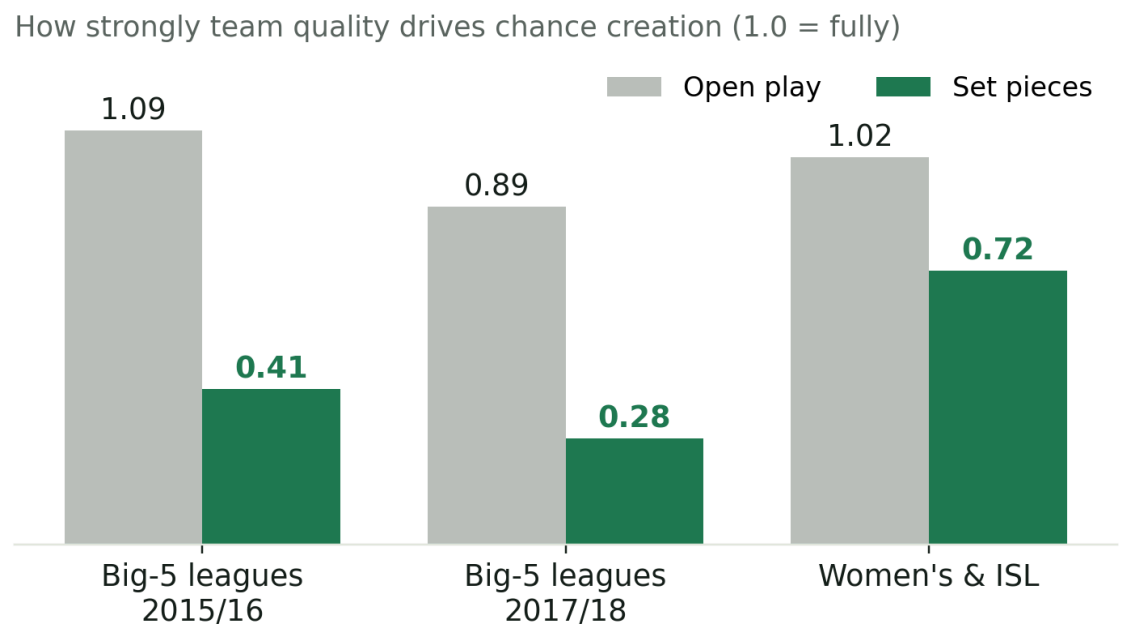


Figure 3. Elasticity of chance creation with respect to team strength (1.0 = rises one-for-one). Set pieces in green. Three independent datasets.

EVIDENCE

- Set-piece chances depend on quality **0.28–0.72** as strongly as open play (0.89–1.09), in all three datasets.
- In mismatches, underdogs win **37–38%** of set-piece xG but only **33%** of open-play xG.
- Set-piece efficiency is only modestly persistent from one half-season to the next. It has to be coached continuously; it can't simply be bought.
- How many attackers go into the box at corners made **no measurable difference** (round 24 agent, null in all three populations).

IN PRACTICE

Treat set pieces as a main attacking channel, not a break in play. The lever is routine quality and delivery (movement, screens, second-phase positions), not bodies in the box.

WHY IT MATTERS FOR UNDERDOGS

Open-play chance creation is largely bought with player quality. Set-piece quality is largely built on the training ground. It is the channel where coaching time converts most directly into chances for a cheaper squad.

PARTLY CONFIRMED

5 PARTLY TRUE

Three ideas hold up in part. The mechanism is real, but the payoff is not proven or is smaller than believed. These are where a club's own data or a pitch trial could add the missing piece.

5.1 BAIT ONE SIDE TO OPEN THE OTHER



Figure 4. Two or more actions on one side pull the block across, so the far side opens.

TEST	RESULT
Far-lane overload after a bait vs none	+.105 / +.079 / +.052 PASS
Tracking: far-side space after ≥3 s one side	+1.33 m (1.06–1.59) PASS
Block slide per metre the ball moves (tracking)	0.38 m, lag 1.4–1.7 s
Switch rate after bait vs none	7.9% vs 4.8% (intl)
Chances after switch: bait vs no bait	−.0007 / +.0023 / +.0019 FAIL
Tracking: final third within 10 s	49% vs 49% FAIL

Rows give men's intl / men's club / women's. The defending block reacts about 1.5 s late, which is why baiting works.

INTERPRETATION

The bait reliably *creates* space. On average teams don't yet convert that space into more chances. The open question for coaching is the quality and speed of the switch and what follows it.

5.2 PASS INTO THE LANE WHERE YOU OUTNUMBER THEM

TEST (OVERLOAD PASS GENUINELY ON)	RESULT
Completion, overload vs crowded lane	+2.8 / +3.3 / +4.2 pts PASS
Net xG, 20 s (stress-tested)	+.0014 / −.0001 / +.0011 FAIL
Share choosing the crowded lane	44–46% (not 66%)

"Genuinely on" means the receiver is visible, 5–35 yds away, has no opponent within 3 yds, and has a clear passing line. When the option is really there, players already take it more than half the time.

5.3 COUNT THE DEFENDERS WHEN YOU WIN THE BALL

At the moment of a regain, count the opponents between the ball and their goal. It is one of the most reliable readings in the data.

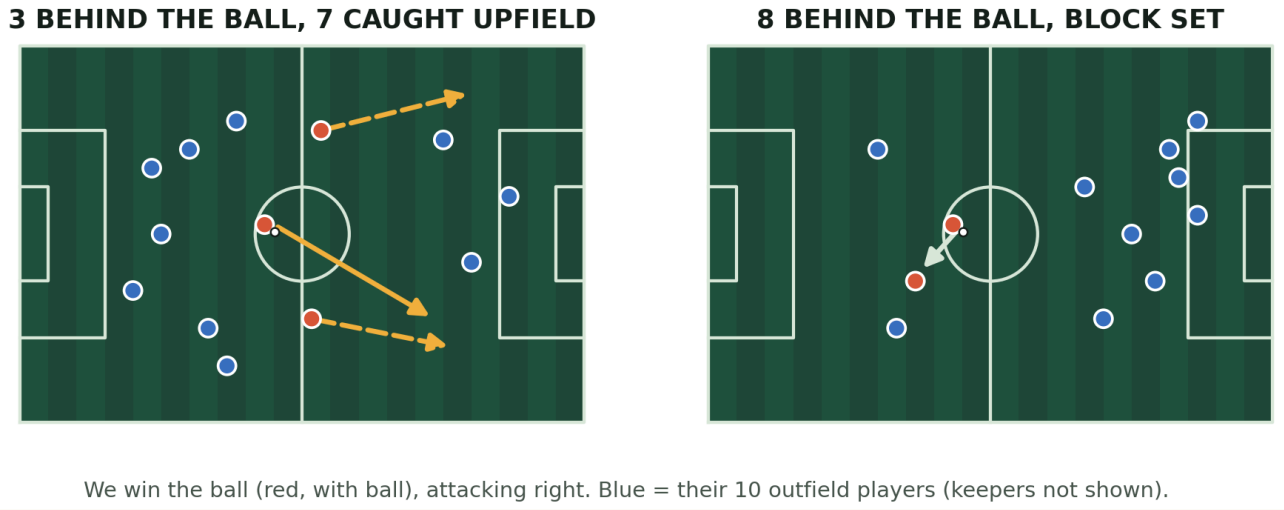


Figure 5. The same regain with 3 behind the ball (left) and 8 (right). Ten outfield opponents in each.

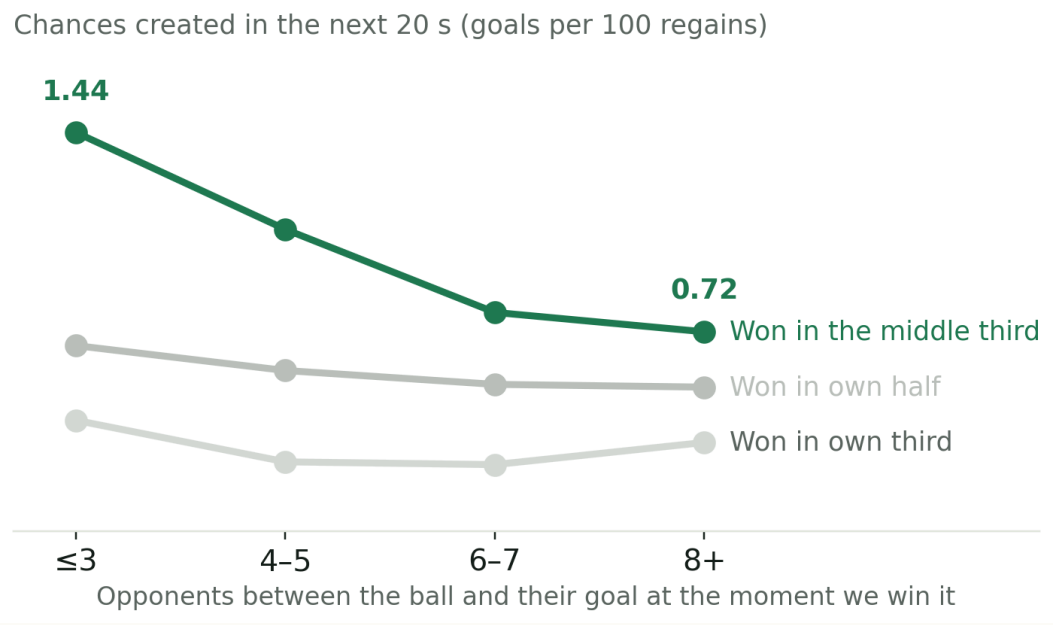


Figure 6. Chances in the 20 s after a regain, by opponents behind the ball. 18,158 regains.

TEST	MEN'S INTL	MEN'S CLUB	WOMEN'S	VERDICT
Value multiplier per extra opponent behind the ball	×0.81	×0.89	×0.88	PASS
Going fast vs keeping it, when ≤3 back (stress-tested, net xG)	+.018	+.009	+.003	UNDERPOWERED
Extra runners committed, once the count is known	×1.09	×1.00	×1.06	NO EFFECT

INTERPRETATION

The count tells you *when* a counter is worth more, reliably and in every population. The estimates for "go fast" point the right way in all three populations, but there are too few such moments to prove it beats keeping the ball. Treat "count" as a reading habit, not a rule.

6 BELIEFS THAT FAILED

Each got the same procedure. "No edge" means no measurable advantage across thousands of matches. It does not prove the idea never works for a particular team.

BELIEF	WHAT WE FOUND	DATA
"Most vulnerable right after scoring"	Goals conceded in the 5 min after scoring were 0.59–0.79× normal (fewer, because of the kick-off and celebration)	4 datasets
Press on triggers (back pass, touchline, bad touch, long ball, crowded midfield)	No trigger made pressing pay more	160k receipts
Press more, everywhere	1 of 102 pressing situations showed a gain, and it later failed the stress test	143k
Take free kicks and throws quickly	0.5–0.8 fewer defenders goal-side, but no better chances (+.0018 … +.0003)	160k restarts
Attack the opponent's weak side	Weakness profiles don't repeat; exploiting them: −.046 / −.026 / −.012	~4,000 matches
Pick the formation for the matchup	Out-of-sample gain negative in all three datasets	3 datasets
Early or attacking subs when losing	No benefit after adjusting for momentum and strength	3 datasets
Chase earlier when losing	Late-game patterns reflect the leader sitting back, not the chaser's choice	EPL, 20 yrs
More runners on the counter	Multiplier ×1.09 / ×1.00 / ×1.06, not significant	18k regains
"Only cross with bodies in the box"	Crossing beats carrying from wide either way; no dose-response with box numbers	12.5k
Overload the six-yard box at corners	No effect of box numbers (+.004 pooled, ns)	1,345 corners
"Go forward more" as a team style	28 of 28 habit tests null	3 datasets
Play faster passing football	Positive at first, then vanished once team quality was controlled (−0.03)	3 datasets
Adapt to strict referees	No effect	EPL, 20 yrs
Tactics only underdogs should use	369 underdog-vs-favourite interaction tests: none	360
Carry into space rather than dribble	+.019 in men's internationals; +.004 elsewhere, ns	360
Fast, flat long balls	+6–11 pts like-for-like; vanished under the stress test	33k balls

ILLUSION

7 WHY "SHOOT MORE" IS AN ILLUSION

Near the box, shooting looks far better than passing back, even like-for-like. It was the headline of the first version of this work. It does not survive.

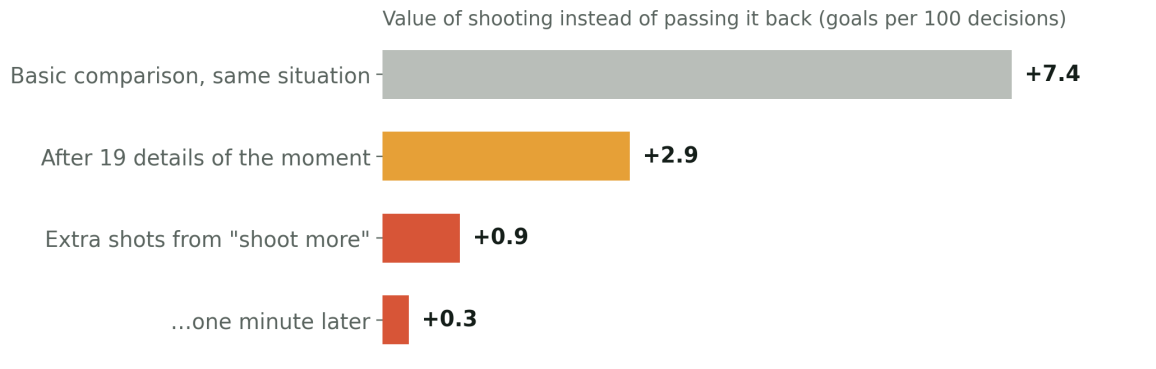


Figure 7. Value of shooting instead of recycling near the box, as the comparison gets stricter. Averages of the three populations; 16,751 shots and recycles.

CHECK	MEN'S INTL	MEN'S CLUB	WOMEN'S	VERDICT
Like-for-like, with keeper position	+.071	+.089	+.068	PASS
Doubly robust, 19 details	+.029	+.027	+.031	SHRINKS
Extra shots a coach would add	+.009	+.008	+.010	TINY
... measured 60 s later	+.002	+.004	+.003	ZERO
Placebo: 20 s before the shot	+.026	+.027	+.021	FAILS
Where players rarely shoot	+.000	+.000	+.005	ZERO

WHAT'S HAPPENING

Players shoot when the shot is on: keeper out of position, clean contact, a scramble. The shot inherits a moment that was already good, so the "gain" exists before the decision.

THE GENERAL LESSON

Any analysis that sees less than the player will credit the action instead of the moment. The placebo question (*was the moment before already different?*) also removed "carry into space" and "press the exposed receiver".

8 CAN A SIMULATOR INVENT TACTICS?

Real data can only judge things players already do. To look for something genuinely new, we built a 22-player match simulator and used it to invent and test tactics.

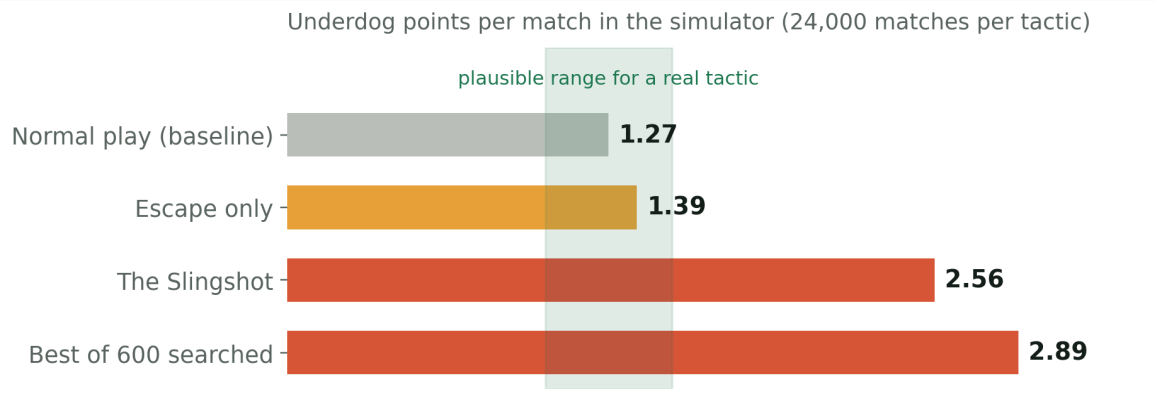


Figure 8. Points per match for a weaker team in the simulator. The shaded band is roughly what a real tactic could plausibly add.

WHAT WE TRIED

- **The Slingshot**, an invented tactic: a lopsided shape, an isolated far winger, release after a bait, and long balls only to +1. It "won" 83% of matches.
- **Grid search** over 3,000 tactic combinations × 100 matches, the best re-tested on fresh matches, 5 opponent styles and 6–8 physics variants.
- **Calibration** to real football: goals, shots, passing, tackles, the counter gradient, and the tracking-data block slide and lag.

WHAT WE LEARNED

- Every top tactic was an **exploit** of the simulator's defending: never press, never mark the far side.
- Even calibrated to real averages, the simulator could not reproduce real-world nulls. Pressing swung results by ~1.9 points a match; real data shows ≈ 0.
- **Conclusion:** agent simulators react 10–50× more to tactics than real football does. They are useful for generating ideas, but useless as evidence.

ONE IDEA THAT AGREED EVERYWHERE

"Escape" (go long only where you have +1, with a runner arriving) was positive in every simulator variant and is backed by both real datasets. That is the long-ball rule of Section 3.

9 RECOMMENDATIONS

FOR COACHES

DO	HOW
Make long balls 2 v 1	Name a support runner for every goal kick and long ball. In training, score long-ball games on retention after 5 s, with a bonus only for +1 (not +2).
Build set pieces as a channel	Regular, timed set-piece blocks. Measure delivery and second-phase positions, not how many players attack the box.
Use the bait deliberately	Positional games where the switch is only "live" after 2+ actions on one side. Work on the speed and quality of what follows the switch.
Teach "count" as a reading	On every regain, players call the number of opponents behind the ball. Fewer means the counter is worth more.
Stop over-investing	In pressing triggers, "weak side" plans and early-sub rules. Our data shows no return from them.

FOR ANALYSTS

1. Compare choices only like-for-like (same situation), never raw.
2. Run the placebo: if the moment before the decision already differs, the stat describes the moment.
3. Require replication in a second population before briefing staff.
4. Treat simulator or "AI tactics engine" outputs as hypotheses, never as evidence.

FOR CLUBS: A LOW-COST PILOT

EIGHT-WEEK SECOND-BALL PILOT

Weeks 1–2: measure the local balance and 5-s retention of every long ball from your own video or tracking. Weeks 3–6: coach "one more where it lands". Weeks 7–8: re-measure. The expected signal is about 15–20 points of retention on balls that reach +1, large enough to see within a season.

10 LIMITATIONS AND NEXT STEPS

LIMITATIONS

- **Observational data.** Even stress-tested estimates can miss things players see, such as body shape. The final proof is a pitch trial.
- **Snapshots, not movement.** 360 data shows positions at each event, and only inside the camera's view. Tracking covers just 30 matches.
- **Short windows.** Effects are measured over 5–60 s; longer-run effects (fatigue, opponent adaptation) are not captured.
- **Average effects.** A rule that is null on average may still work for a particular team or player.
- **Earlier overclaims.** Versions 1–3 of this work reported rules that later failed stricter tests. All are documented and corrected publicly.

LOCKED PREDICTIONS

Ten numeric predictions were committed publicly (14–16 October 2026) before the next StatsBomb 360 release. One command tests them, and the outcome will be published whatever it is.

NEXT STEPS

- Season-scale tracking data, to test off-ball movement and the bait's payoff properly.
- A club pilot of the long-ball rule (Section 9).
- Team-specific testing: which beliefs hold for one club's own style.
- Turning the method into an open "belief tester" any analyst can run.

BOTTOM LINE

Football has fewer magic tactics than it believes. The real edges are small and practical, and they sit where teams pay least attention: long balls and set pieces. Most of the value of rigorous testing is in what it lets a club stop doing.

A TEST RECORD AND SOURCES

ROUNDS	WHAT WAS TESTED	OUTCOME
1–10	Game-state tempo, style knobs, chance channels, underdog value, counter windows	Set-piece elasticity; most style effects null
11–12	Automated tactic miner (6,847 rules); habit tests (28)	0 pass false-discovery control; 28/28 null
13–15	360 situation miner (771 + 997 + 102 rules); counter trigger	Count confirmed; shot rules found
16–21	Possession simulator, weak spots, overload, formations, sweet spots, restarts, vulnerability windows	Overload found; most null; myths busted
Agents 1–2	Shooter roles, combinations, crossing, restarts, triggers, subs, second ball, runners, corners, keeper	Second ball confirmed; most null
22	Stress tests (doubly robust + placebo) on every rule; underdog-only mining	Strike, Run, Press killed; no underdog-only rules
23	Overload availability, bait-and-switch, long-ball chain	Partial results (Section 5)
24–27	Invented tactic, simulator grid search and calibration, tracking data, fast long balls	Simulator exploits; long-ball rule replicated

REFERENCES AND DATA

- StatsBomb Open Data, including 360 freeze frames. Used under the StatsBomb public data user agreement.
- Pappalardo, L. et al. (2019). A public data set of spatio-temporal match events in soccer competitions. *Scientific Data* 6, 236.
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Every pre-registration, deviation, script and result file is public at **github.com/ranausmanai/count-and-strike**, including all failed tests.

ABOUT THE AUTHOR

RANA MUHAMMAD USMAN

An independent football analyst. Project Underdog started with a single question: what can a weaker team do, in the moment, to gain a real edge over a stronger one? This paper is the honest answer so far, built from public data, every test registered in advance, and every failure published.

SERVICES FOR CLUBS

- Second-ball and long-ball audit from your own data
- Testing your staff's tactical beliefs with this method
- Set-piece and opponent review for the weekly cycle
- Design and analysis of in-season pilots

CONTACT

github.com/ranausmanai/count-and-strike